UNIVERSITY OF THE PUNJAB

BS (After Associate Degree) Sixth Semester – Fall 2023

Paper: Mathematical Methods of Physics-II (Common with BS4Y 6th)

Course Code: PHYS-3503

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Solve the following: (6x5=30)

1) By using the Levi–Civita symbol, evaluate the determinant of the given matrix

$$\begin{pmatrix} 2 & 1 & -3 \\ 3 & 4 & 0 \\ 1 & -2 & 1 \end{pmatrix}$$

3 2) Any arbitrary tensor of rank two can be expressed as a sum of symmetric and anti-symmetric tensor of rank two. 1

5 3) Show that every cyclic group is abelian 5

5 4) Show that the set S = {1, 3, 5, 7} is a group under multiplication (mod 8). 3

5 5) Expand $f(z) = \frac{1}{z(z-1)}$ in a Laurent series valid for the annular domain $1 < |z|$.

6) By using Cauchy's residue theorem to evaluate the given integral along the indicated contour

$$\oint_c \frac{\tan z}{z}\, dz$$

where C: $|z - 1| = 2$.

Q.2. Solve the following. (5x6=30)

1) Evaluate the Cauchy principal value of

$$\int_{-\infty}^{\infty} \frac{1}{x^4 + 1} dx$$

2) By using Residue theorem, evaluate the integral

$$\int_0^{2\pi} \frac{d\theta}{1 + a\cos\theta}$$

where $|a| < 1$.

4 3) Show that the T_{ij} given by $T = [T_{ij}] = \begin{pmatrix} x_2^2 & -x_1x_2 \\ -x_1x_2 & x_1^2 \end{pmatrix}$ are the components of a second order tensor. 3

4) Show that the quantities $g_{ij} = e_i \cdot e_j$ form the covariant components of a second-order tensor.

5 5) If every element of a group G is its own inverse, show that G is abelian.

6

UNIVERSITY OF THE PUNJAB

BS (After Associate Degree) : Sixth Semester – Spring 2023

Roll No. [redacted]

Paper: Mathematical Methods of Physics-II   Course Code: PHYS-3503   Time: 3 Hrs.   Marks: 6

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answer of the following questions: (6x5=30)

1. What is Kronecker delta? Prove that $\varepsilon_{ijk}\varepsilon_{ijk} = 6$.
2. Apply tensor to show that $\vec{a} \bullet \vec{b} \times \vec{c} = \vec{b} \bullet \vec{c} \times \vec{a} - \vec{c} \bullet \vec{a} \times \vec{b}$.
3. Prove that a finite group of prime order is always cyslic.
4. Show that an infinite cyclic group has exactly two different generators.
5. Evaluate the integral $\int_C \frac{e^{2z} + SinZ}{(Z \mp 1)^4}\, dZ$ where encloses the point $Z = -1$.
6. A function $f(Z)$ which has no singularity in the finite part of the plane or at infinity is a constant.

Q.2. Solve the following. (5x6=30)

1. Show that the transformation from X to X' frame is
$$X_1' = \frac{1}{3}(2X_1 + 2X_2 - X_3), \quad X_2' = \frac{1}{3}(2X_1 - X_2 + 2X_3)$$
$$X_3' = \frac{1}{3}(-X_1 + 2X_2 + 2X_3)$$
is orthogonal . Also prove that $A_{ii}' = A_{pp}$.

2. State and prove Lagrange's theorem. Also define Christoffel symbols.

3. Show that $Aut(Z_n)$ is isomorphic to $U(n)$ where $n \in Z^+$.

4. Evaluate $\int_0^{\infty} \frac{1 + 2Cos\theta}{5 + 4Cos\theta}\, d\theta$.

5. State and prove Cauchy's Residue Theorem.

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – Spring 2023

Roll N

Paper: Mathematical Methods of Physics-II **(RE-CONDUCT)**
Course Code: PHYS-3503

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Solve the following: (6x5=30)

1. What is dummy index? Expand $a_{pqr} X_p X_{qq}$.
2. Show that the product of a tensor and a Pseudo tensor is also a Pseudo tensor.
3. State and prove Lagrange theorem.
4. Prove that every cyclic group is Abelian.
5. Prove that the function $f(Z) = \dfrac{x^3(1+i) - y^3(1-i)}{x^2+y^2} \quad when \quad Z \neq 0$

$$= 0 \quad when \quad Z = 0$$

6. Expand $\log(1+Z)$ in a Taylor's series about $Z = 0$ and determine the region of convergence for the resulting series.

Q.2. Solve the following. (5x6=30)

1. The transformation from X to X' frame is $X_1' = -\frac{3}{7}X_1 - \frac{6}{7}X_2 - \frac{2}{7}X_3$, $X_2' = -\frac{2}{7}X_1 + \frac{3}{7}X_2 - \frac{6}{7}X_3$, $X_3' = \frac{6}{7}X_1 - \frac{2}{7}X_2 - \frac{3}{7}X_3$. Find the components of of 3^{rd} rank tensor where all components (except $A_{111} = A_{222} = 1$ and $A_{212} = -2$) are zero.
2. Show that $A_{pq} = \begin{pmatrix} x_2^2 & x_1x_2 \\ -x_1x_2 & x_1^2 \end{pmatrix}$ are not the components of a 2^{nd} order tensor.
3. Prove that the set of all automorphisms of a group G is a group under the operation of composition of functions
4. Find the Residues of $f(Z) = \dfrac{e^Z}{Z^2(Z-\pi i)^4}$.
5. Evaluate $\int_0^\infty \dfrac{Cos\ mx}{a^2+x^2}\,dx$ and deduce the value of $\int_0^\infty \dfrac{x\ Sin\ mx}{a^2+x^2}\,dx$ where a and m are constants.

UNIVERSITY OF THE PUNJAB

Roll

BS (After Associate Degree) : Sixth Semester – Spring 2022

Paper: Mathematical Methods of Physics-II | Course Code: PHYS-3503 | Time: 3 Hrs. | Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answer of the following questions: (6x5=30)

1. What is dummy index? Expand $a_{pq}X_pX_q$.
2. Show that the product of a tensor and a Pseudo tensor is also a Pseudo tensor.
3. State and prove Lagrange theorem.
4. Prove that every cyclic group is Abelian.
5. Prove that the function $$f(Z)=\frac{x^3(1+i)-y^3(1-i)}{x^2+y^2} \quad when \quad Z\neq 0$$ $$=0 \quad when \quad Z=0$$
6. Expand $\log(1+Z)$ in a Taylor's series about $Z=0$ and determine the region of convergence for the resulting series.

Q.2. Solve the following. (5x6=30)

1. The transformatio from X to X' frame is $X_1'=-\frac{3}{7}X_1-\frac{6}{7}X_2-\frac{2}{7}X_3$, $X_2'=-\frac{2}{7}X_1+\frac{3}{7}X_2-\frac{6}{7}X_3$, $X_3'=\frac{6}{7}X_1-\frac{2}{7}X_2-\frac{3}{7}X_3$. Find the components of of 3^{rd} rank tensor where all components (except $A_{111}=A_{222}=1$ and $A_{212}=-2$) are zero.
2. Show that $A_{pq}=\begin{pmatrix} x_2^2 & x_1x_2 \\ -x_1x_2 & x_1^2 \end{pmatrix}$ are not the components of a 2^{nd} order tensor.
3. Prove that the se of all automorphisms of a group G is a group under the operation of comp sition of functions
4. Find the F esidues of $f(Z)=\dfrac{e^Z}{Z^2(Z-\pi i)^4}$.
5. Evaluate $\int_0^\infty \frac{\cos nx}{a^2+x^2}dx$ and deduce the value of $\int_0^\infty \frac{x \sin mx}{a^2+x^2}dx$ where a and m are constants.

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : Fifth Semester – Fall 2021

Roll No. ....

Paper: Mathematical Methods of Physics | Course Code: PHY-302 | Time: 3 Hrs. | Marks: 60

Q.1. Solve the following: **(6x5=30)**

1. With $\hat{\mathbf{e}}_1$ a unit vector in the direction of increasing u_1, show that

$$\nabla.\hat{\mathbf{e}}_1 = \frac{1}{h_1 h_2 h_3}\frac{\partial}{\partial u_1}(h_2 h_3)$$

2. Show that $\mathbf{r} = \rho\hat{\mathbf{e}}_\rho + z\hat{\mathbf{e}}_z$. Working entirely in circular cylinderical coordinates also prove that $\nabla.\mathbf{r} = 3$ and $\nabla \times \mathbf{r} = 0$. (Note: $x = \rho\cos\phi$, $y = \rho\sin\phi$, $z = z$)

3. Given $A_k = \frac{1}{2}\varepsilon_{ijk}B^{ij}$, with $B^{ij} = -B^{ji}$, antisymmetric tensor, show that

$$B^{mn} = \varepsilon^{mnk}A_k.$$

4. Show that

$$\frac{1}{2\pi i}\oint_C z^{m-n-1}dz, \qquad m \text{ and } n \text{ integers}$$

(with the contour encircling the origin once counterclockwise) is a representation of the Kronecker δ_{mn}.

5. Evaluate

$$\oint_{|z-2i|=4}\frac{z}{z^2+9}dz,$$

by using Cauchy's integral formula.

6. Use the divergence theorem to establish Green's identity

$$\iint_S (f\nabla g).\mathbf{n}dS = \iiint_D (f\nabla^2 g + \nabla f.\nabla g)\,dV,$$

here f and g are scalar functions with continuous second partial derivatives.

Q.2. Solve the following. **(5x6=30)**

1. Evaluate the Cauchy principal value of

$$\int_{-\infty}^{+\infty}\frac{dx}{(x^2+a^2)^2}(a>0).$$

2. The derivative of the second-order tensor $\mathbf{T}$ with respect to the coordinate u^k, find an expression for the covariant derivative $T^{ij}_{;k}$ of its contravariant components.

3. Calculate the elements g_{ij} of the metric tensor for spherical polar coordinate. Also calculate the Γ^k_{ij} for spherical polar coordinates. (Note: $x = r\cos\phi\sin\theta$, $y = \rho\sin\phi\sin\theta$, $z = r\cos\theta$)

4. Use the Cauchy's residue theorem to evaluate

$$\oint_{|z-1|=2}\frac{\tan z}{z}dz.$$

5. Find the spherical coordinate components of the velocity and acceleration of a moving particle. ($x = r\cos\phi\sin\theta$, $y = \rho\sin\phi\sin\theta$, $z = r\cos\theta$)

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : Fifth Semester – 2020

R.

Paper: Mathematical Methods of Physics-
Course Code: PHY-302 **Part – II**

Time: 2 Hrs. 45 Min. **Marks: 50**

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Solve the following: **(5x4=20)**

1. Show that an arbitrary tensor (neither symmetric nor antisymmetric) can always be written as sum of a symmetric tensor and an antisymmetric tensor.
2. Evaluate the surface integral $I = \int_S \mathbf{a}.d\mathbf{S}$, where $\mathbf{a} = x\mathbf{i}$ and S is the surface of the hemisphere $x^2 + y^2 + z^2 = a^2$ with $z \geq 0$.
3. Show that the quantities $g_{ij} = \mathbf{e}_i.\mathbf{e}_j$ form the covariant components of a second-rank tensor.
4. Determine the order of the poles of $f(z) = \frac{\sin z}{z^3 - z}$.
5. Expand $f(z) = \frac{1}{z(z-1)}$ in a Laurent series valid for $1 < |z|$.

Q.3. Solve the following. **(5×6=30)**

1. Evaluate the Cauchy principal value of

$$\int_{-\infty}^{+\infty} \frac{x \sin x}{x(x^2+2)} dx.$$

2. Prove the following identities

$$(\epsilon^i.\epsilon^j)(\epsilon_j.\epsilon_k) = \delta^i_k, \qquad \Gamma^m_{jk} = \Gamma^m_{kj}.$$

3. Calculate the basis vectors $\mathbf{e}_r, \mathbf{e}_\theta$, and $\mathbf{e}_\phi$. Also compute the basis vectors $\epsilon_r, \epsilon_\theta$ and ϵ_ϕ. FurtherShow that $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_\phi\}$ and $\{\epsilon_r, \epsilon_\theta, \epsilon_\phi\}$ are reciprocal systems of vectors. ($x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$ and $z = r\cos\theta$).
4. Use an indented contour and residues, show that

$$\text{P.V}\left(\int_{-\infty}^{+\infty} \frac{\sin x}{x} dx\right) = \pi.$$

5. Suppose that C is a piecewise-smooth simple closed curve bounding a simple connected region R. If the functions $P(x,y)$ and $Q(x,y)$ and their partial derivatives are continuous on R, then show that

$$\oint_C (P(x,y)dx + Q(x,y)dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA.$$

UNIVERSITY OF THE PUNJAB

Fifth Semester – 2019

Examination: B.S. 4 Years Program

Roll No. 219599....

PAPER: Mathematical Methods of Physics-.

Course Code: PHY-302 Part – II

MAX. TIME: 2 Hrs. 45 Min.

MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Questions with Short Answers. **(5x4=20)**

1. If $\hat{e}_1$ a unit vector in the direction of increasing u_1, show that

$$\nabla.\hat{e}_1 = \frac{1}{h_1h_2h_3}\frac{\partial(h_2h_3)}{\partial u_1}.$$

2. Use the divergence theorem to establish following identities

$$\iint_S (f\nabla g).\mathbf{n}dS = \iiint_D (f\nabla^2 g + \nabla f.\nabla g)\,dV.$$

here f and g are scalar functions with continuous second partial derivatives..

3. Given $A_k = \frac{1}{2}\varepsilon_{ijk}B^{ij}$, with $B^{ij} = -B^{ji}$, antisymmetric tensor, show that

$$B^{mn} = \varepsilon^{mnk}A_k.$$

4. Determine the order of the poles of $f(z) = \frac{\cot(\pi z)}{z^2}$.

5. Suppose z_0 is any constant complex number interior to any simple closed contour C. Show that

$$\oint_C \frac{dz}{(z-z_0)^n} = \begin{cases} 2\pi i, & n = 1 \\ 0, & n \text{ is a positive integer} \neq 1 \end{cases}$$

Q.3. Questions with Brief Answers. **(5x6=30)**

1. Show that

$$\int_0^\pi \frac{d\theta}{(a+b\cos\theta)^2} = \frac{\pi a}{(a^2-1)^{\frac{3}{2}}}.$$

2. Show that

$$\Gamma^m_{ij} = \frac{1}{2}g^{mk}\left(\frac{\partial g_{jk}}{\partial u^i} + \frac{\partial g_{ki}}{\partial u^j} - \frac{\partial g_{ij}}{\partial u^k}\right).$$

3. Using Cauchy's integral formula for derivatives, evaluate

$$\oint \frac{z+1}{z^4+4z^3}dz,$$

where C is the circle $|z| = 1$.

4. Evaluate

$$\int_{-\infty}^{+\infty} \frac{x^2}{1+x^4}dx.$$

5. Find the spherical coordinate components of the velocity and acceleration of a moving particle. ($x = r\cos\phi\sin\theta$, $y = \rho\sin\phi\sin\theta$, $z = r\cos\theta$)

UNIVERSITY OF THE PUNJAB

Fifth Semester 2018
Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Mathematical Methods of Physics-
Course Code: PHY-302

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

Section-II (Short Questions)

Marks=20

1. Show that $\{\mathbf{e}_i\}$ and $\{\epsilon_i\}$ are reciprocal systems of vectors.
2. Expand $f(z) = e^{3/z}$in a Laurent series valid for $0 < |z|$.
3. Show that $z = 0$ is an essential singularity of $f(z) = z^3 \sin(1/z)$.
4. Show that $\mathbf{r} = \rho\hat{\mathbf{e}}_\rho + z\hat{\mathbf{e}}_z$, also prove that $\nabla.\mathbf{r} = 3$ and $\nabla \times \mathbf{r} = 0$. (Note: $x = \rho\cos\phi$, $y = \rho\sin\phi$, $z = z$)
5. Using the calculus of residues, show that $\int_0^\pi \cos^{2n}\theta d\theta = \pi\frac{(2n)!}{2^{2n}(n!)^2}$, $n = 0, 1, 2....$

Section-III

Marks=30

1. Evaluate
$$\int_0^{2\pi} \frac{\cos(3\theta)d\theta}{5 - 4\cos\theta}.$$
2. Evaluate
$$\oint_{|z-2i|=4} \frac{z}{z^2+9}dz,$$
by using Cauchy's integral formula.
3. By considering the derivative of the second-order tensor $\mathbf{T}$ with respect to the coordinate u^k, find an expression for the covariant derivative $T_{ij;k}$ of its contravariant components.
4. The electric field $\mathbf{E} = -\nabla\varphi$; this is derived from a scalar, the electrostatic potential φ, and has components $E_i = -\frac{\partial\varphi}{\partial x_i}$. Show that $\mathbf{E}$ is a first order tensor.
5. Find the circular cylindrical components of the velocity and acceleration of a moving particle. (Hint: $\mathbf{r}(t) = \rho(t)\hat{\mathbf{e}}_\rho(t) + z(t)\hat{\mathbf{e}}_z$ and $x = \rho\cos\phi$, $y = \rho\sin\phi$, $z = z$)

UNIVERSITY OF THE PUNJAB

Fifth Semester 2017
Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Mathematical Methods of Physics-
Course Code: PHY-302

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

Section-II (Short Questions) Marks=20

1. If $\mathbf{B} = \nabla \times \mathbf{A}$, show that
$$\oint \mathbf{B}.d\sigma = 0,$$
for any closed surface S.

2. Expand
$$f(z) = \frac{1}{z(1-z)^2}$$
in a Laurent series valid for $0 < |z| < 1$.

3. Determine whether $z = 0$ is an isolated or non-isolated singularity of $f(z) = \tan(1/z)$.

4. A vector $\mathbf{A}$ is decomposed into a radial vector $\mathbf{A}_r$ and a tangential vector $\mathbf{A}_t$. If $\hat{\mathbf{r}}$ is a unit vector in the radial direction, show that
$$\begin{aligned} \mathbf{A}_r &= \hat{\mathbf{r}}(\mathbf{A}.\hat{\mathbf{r}}), \\ \mathbf{A}_t &= -\hat{\mathbf{r}}\times(\hat{\mathbf{r}} \times \mathbf{A}). \end{aligned}$$

5. Evaluate $\oint_C \frac{z+1}{z^4+4z^3}dz$, where C is the circle $|z| = 1$.

Section-III Marks=30

1. Show that
$$\int_0^\pi \frac{d\theta}{(a+\cos\theta)^2} = \frac{a\pi}{\left(\sqrt{a^2-1}\right)^3}, \qquad (a > 1).$$

2. Use Cauchy's residue theorem to evaluate
$$\oint_{|z|=\frac{1}{2}} \frac{\cot \pi z}{z^2} dz.$$

3. Suppose $A = [a_{ij}]$, $B = [b^{ij}]$ and that $B = A^{-1}$. By considering the determinant $a = |A|$, show that
$$\frac{\partial a}{\partial u^k} = ab^{ji}\frac{\partial a_{ij}}{\partial u^k}.$$

4. We may define Christoffel symbols of the first kind by
$$\Gamma_{ijk} = g_{il}\Gamma^l_{jk}.$$
Show that these are given by
$$\Gamma_{kij} = \frac{1}{2}\left\{\frac{\partial g_{ik}}{\partial u^j} + \frac{\partial g_{jk}}{\partial u^i} - \frac{\partial g_{ij}}{\partial u^k}\right\}.$$

5. Prove the expression $\nabla.\mathbf{a} = \frac{1}{h_1h_2h_3}\left[\frac{\partial}{\partial u_1}(h_2h_3a_1) + \frac{\partial}{\partial u_2}(h_1h_3a_2) + \frac{\partial}{\partial u_3}(h_1h_2a_3)\right]$ in orthogonal curvi linear coordinates.

UNIVERSITY OF THE PUNJAB

Fifth Semester 2016
Examination: B.S. 4 Years Programme

Roll No. 009583

PAPER: Mathematical Methods of Physics-
Course Code: PHY-302

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

Section-II (Short Questions) Marks=20

1. Use the divergence theorem to establish Green's identity

$$\iint_S (f\nabla g).ndS = \iiint_D (f\nabla^2 g + \nabla f.\nabla g)\,dV,$$

where f and g are scalar functions with continuous second partial derivatives.

2. Expand

$$f(z) = \frac{z}{(z+1)(z-2)}$$

in a Laurent series valid for $0 < |z-2| < 3$.

3. With $\hat{e}_1$ a unit vector in the direction of increasing u_1, show that

$$\nabla.\hat{e}_1 = \frac{1}{h_1h_2h_3}\frac{\partial}{\partial u_1}(h_2h_3)$$

4. If $\mathbf{A}$ is irrotational (i.e. $\nabla\times\mathbf{A}=0$), show that $\mathbf{A}\times\mathbf{r}$ is solenoidal (i.e. $\nabla.(\mathbf{A}\times\mathbf{r})=0$).

5. Show whether or not the function $f(z) = \mathrm{Re}(z)$ is analytic.

Section-III Marks=30

1. Show that

$$\int_0^{2\pi}\frac{d\theta}{a+b\cos\theta} = \frac{2\pi}{\sqrt{a^2-b^2}}, \qquad (a>b>0).$$

2. Use the Cauchy's residue theorem to evaluate

$$\oint_{|z-1|=2}\frac{\tan z}{z}dz.$$

3. Find the circular cylindrical polar components of the velocity and acceleration of a moving particle. $(x=\rho\cos\phi,\ y=\rho\sin\phi,\ z=z)$

4. Calculate the elements g_{ij} of the metric tensor for spherical polar coordinate. $(x=r\sin\theta\cos\phi,\ y=r\sin\theta\sin\phi,\ z=r\cos\theta)$

5. By considering the derivative of the second-order tensor $\mathbf{T}$ with respect to the coordinate u^k, find an expression for the covariant derivative $T^{ij}_{;k}$ of its contravariant components.

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PHY

UNIVERSITY OF THE PUNJAB

Fifth Semester 2015

Examination: B.S. 4 Years Programme

Roll No. 9922

PAPER: Mathematical Methods of Physics-
Course Code: PHY-302

TIME ALLOWED: 2 hrs. & 30 mi
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

Section-II (Short Questions)

Marks=20

1. The double summation $K_{ij}A^iB^j$ is invariant for any two vectors A^i and B^j. Prove that K_{ij} is a second rank tensor.

2. Expand

$$f(z) = \frac{1}{(z-1)^2(z-3)}$$

in a Laurent series valid for $0 < |z-3| < 2$.

3. If a vector function depends on both space coordinates (x, y, z) and time t, (that is $\mathbf{F} = \mathbf{F}(x, y, z, t)$), show that

$$d\mathbf{F} = (d\mathbf{r}.\nabla)\mathbf{F} + \frac{\partial \mathbf{F}}{\partial t}dt$$

4. Show that the covariant derivative of the metric tensor is zero (i.e. $g_{ij;k} = g^{ij}_{;k} = 0$).

5. If there is some common region in which $w_1 = u(x, y) + iv(x, y)$ and $w_2 = w_1^* = u(x, y) - iv(x, y)$ are both analytic, prove that both $u(x, y)$ and $v(x, y)$ are constants.

Section-III

Marks=30

1. Show that

$$\int_0^{2\pi} \frac{\sin^2\theta}{a + b\cos\theta}d\theta = \frac{2\pi}{b^2}\left(a - \sqrt{a^2 - b^2}\right). \qquad (a > b > 0).$$

2. Use the Cauchy's residue theorem to evaluate

$$\int_{|z|=3} \frac{e^z}{z^3 + 2z^2}dz.$$

3. Find the spherical polar coordinates components of the velocity of a moving particle. ($x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta$)

4. Show that, in general coordinates, the quantities $\frac{\partial x^i}{\partial q^j}$ do not form the components of a tensor.

5. Show that $\{\mathbf{e}_i\}$ and $\{\epsilon_i\}$ are reciprocal system of vectors. => Page (8) PU-notes

UNIVERSITY OF THE PUNJAB

Fifth Semester 2014
Examination: B.S. 4 Years Programme

Roll No. 018322

M.Ch

PAPER: Mathematical Methods of Physics
Course Code: PHY-302

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

Section-II (Short Questions)

Marks=20

1. If v_i are the components of a first-order tensor, show that $\nabla.\mathbf{v} = \frac{\partial v_i}{\partial x_i}$ is a zero-order tensor.
2. Expand

$$f(z) = \frac{z - \sin z}{z^5}$$

464 (Bin Amin)

in a Laurent series valid for $0 < |z|$.

3. Use the divergence theorem to establish Green's identity

$$\iint_S (f\nabla g).\mathbf{n}dS = \iiint_D (f\nabla^2 g + \nabla f.\nabla g)\,dV,$$

vector 304 (Bin Amin)

here f and g are scalar functions with continuous second partial derivatives.

4. Show that covariant derivative of a covariant vector V_i is given by

$$V_{i;j} = \frac{\partial V_i}{\partial q^j} - V_k\Gamma^k_{ij}, \text{ where Christoffel symbol } \boxed{\Gamma^k_{ij} = \varepsilon^k.\frac{\partial}{\partial q^i}\varepsilon_j}.$$

5. Show whether or not the function $f(z) = \mathrm{Re}(z)$ is analytic.

Section-III

Marks=30

1. Evaluate the Cauchy principal value of

$$\int_{-\infty}^{+\infty} \frac{x\sin ax\,dx}{x^2+9}\,(a \geq 0).$$

2. Use the Cauchy's residue theorem to evaluate

$$\oint_{|z-1|=2} \frac{\tan z}{z}dz.$$

3. Find the circular cylinderical components of the velocity and acceleration of a moving particle. vector 2-4-8 (Ali Khan)

4. With $\hat{e}_1$ a unit vector in the direction of increasing $\hat{e}_1$, show that

Joe tut.

$$\nabla.\hat{e}_1 = \frac{1}{h_1h_2h_3}\frac{\partial}{\partial q_1}(h_2h_3), \qquad \nabla\times\hat{e}_1 = \frac{1}{h_1}\left[\frac{\hat{e}_2}{h_3}\frac{\partial}{\partial q_3}(h_1) - \frac{\hat{e}_3}{h_2}\frac{\partial}{\partial q_2}(h_1)\right]$$

5. If $T_{i_1 i_2 \cdots i_n}$ is a tensor of rank n, show that $\frac{\partial T_{i_1 i_2 \cdots i_n}}{\partial x^{i_k}}$ is a tensor of rank $n+1$ (Cartesian coordinates). 379 (Bin Amin)

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UNIVERSITY OF THE PUNJAB

Fifth Semester 2013
Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Mathematical Methods of Physics-
Course Code: PHY-302

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

$\frac{1}{1+x}$

Attempt this Paper on Separate Answer Sheet provided.

Section-II (Short Questions)

$\frac{1}{2\pi i}\oint_C \frac{z^m}{z^{(n+1)}}dz$

Marks=20

6.4.2
1.4.1 Pg#429 (MMP) by Arfken

1. Show that

$$\frac{1}{2\pi i}\oint_C z^{m-n-1}dz,$$

m and n integers

z^{m-n-1}

$z^m \cdot z^{-(n+1)}$

(with the contour encircling the origin once counterclockwise) is a representation of the Kronecker δ_{mn}.

2. Expand

$$\boxed{f(z) = \frac{z}{(z+1)(z-2)}}$$

Q.20. Ex.19.3, Pg# 835, CH#19

$= \frac{A}{z+1} + \frac{B}{z-2}$

in a Laurent series valid for the annular domain $0 < |z-2| < 3$.

3. Use the divergence theorem to establish Green's identity

$= \frac{1/3}{z+1} + \frac{2/3}{z-2}$

30.4 (B.in Amm)

$$\iint_S (f\nabla g - g\nabla f).\mathbf{n}\,dS = \iiint_D (f\nabla^2 g - g\nabla^2 f)\,dV,$$

$= \frac{1}{3}\frac{1}{(z-2)+3} = \frac{1}{3}\frac{1}{1+\frac{z-2}{3}}$

here f and g are scalar functions with continuous second partial derivatives.

1.x 4. If $\mathbf{A}$ is irrotational (i.e. $\nabla\times\mathbf{A} = 0$), show that $\mathbf{A}\times\mathbf{r}$ is solenoidal (i.e. $\nabla.(\mathbf{A}\times\mathbf{r}) = 0$). 1.8.3 (Arfken)

5. Verify that

$a\cdot(b\times c) = (a\times b)\cdot c - (a\times c)\cdot b$

$$A^{ik}_{jl} = \delta^i_j\delta^k_l,$$

4.2.4 (a)

is an isotropic fourth-rank tensor.

Section-III

Marks=30

1. Evaluate the Cauchy principal value of

$$\int_{-\infty}^{+\infty}\frac{dx}{(x^2+a^2)^2}\,(a>0).$$

CH.19

2. Evaluate the trigonometric definite integral

$x = a\tan\theta$

$$\int_0^{2\pi}\frac{d\theta}{1+b\cos\theta}\,(|b|<1).$$

CH.19

Pg#843–849

3. ~~Show that the quantities $g_{ij} = \mathbf{e}_i\cdot\mathbf{e}_j$ form the covariant components of a second-rank tensor.~~ Pg#963
4. Show that, in general coordinates, the quantities $\frac{\partial x^i}{\partial q^j}$ do not form the components of a tensor. Pg#968
5. Calculate the elements g_{ij} of the metric tensor for spherical polar coordinates ($x = r\sin\theta\cos\varphi$, $y = r\sin\theta\sin\varphi$, $z = r\cos\theta$). Pg#958

CH#26
(TENSORS)